

Samuel Matthews

SENIOR DATA ENGINEER

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SUMMARY

Results-focused **Senior Data Engineer** with deep expertise across fintech, healthcare, e-commerce, edtech, and consulting. Expert in building and automating cloud-native data pipelines and analytics solutions (AWS, Azure, Python, Spark, SQL, Kafka, Airflow, Databricks). Solid track record in data governance, privacy (HIPAA, GDPR), ETL/ELT, and BI. Trusted to deliver reliable platforms in fast-paced, compliance-driven industries.

SKILLS

- **Programming:** Python, SQL, Java, Scala, Shell Scripting, R
- **Cloud & Data:** AWS (S3, Lambda, Redshift, Glue, EMR), Azure (ADF, Databricks, Synapse, Data Lake), Snowflake, Redshift
- **Big Data:** Apache Spark, Kafka, Hive, Hadoop, Airflow
- **Databases:** SQL Server, MySQL, PostgreSQL, MongoDB, Oracle
- **BI & Visualization:** Power BI, Tableau, Redash
- **DevOps & CI/CD:** Git, Jenkins, Docker, Terraform
- **ETL/ELT:** SSIS, Informatica, Talend, DBT, Azure Data Factory
- **Other:** Pandas, NumPy, Scikit-learn, TensorFlow

WORK EXPERIENCE

Collabera

Jun 2022 – Present

Senior Data Engineer

- Built and automated ETL pipelines using AWS Glue, Redshift, and Airflow for highly regulated financial data, including privacy and compliance requirements (HIPAA, SOX).
- Developed real-time data streaming solutions with Kafka for transaction monitoring and fraud analytics in fintech systems.
- Automated compliance and regulatory reporting workflows using DBT and Python, embedding data quality and governance logic.
- Built machine learning feature stores in Databricks/Spark for credit scoring and risk analysis.
- Implemented cloud data lakehouse architecture with S3, Delta Lake, and Redshift Spectrum, enabling multi-source analytics.
- Configured IAM, encryption, and role-based access controls for secure cloud data infrastructure.
- Migrated healthcare ETL workflows to Python-based cloud pipelines for faster patient reporting.
- Integrated AI-powered document extraction and LLM-based enrichment (OpenAI, Hugging Face) into KYC and claims flows.
- Automated incident monitoring and recovery using Lambda and CloudWatch for always-on data operations.
- Developed metadata management and lineage tracking with DBT and Git.
- Designed RESTful APIs for secure, programmatic access to curated datasets.

- Supported executive dashboards by integrating data with Tableau and Power BI.
- Tuned Spark job configurations and partitioning for cost-effective performance.
- Built unit/integration tests for all ETL code to reduce deployment risk.
- Collaborated with data scientists to enable automated model retraining and inference on streaming data.

Tata Consultancy Services (TCS)

Mar 2018 – May 2022

Senior Data Engineer

- Developed and maintained ETL workflows in Redshift and Glue for payment data, anti-money laundering, and compliance analytics.
- Implemented streaming data pipelines using Kafka and Airflow for real-time KYC and PCI DSS processes.
- Automated transaction data ingestion and batch/streaming workflows from APIs and banking partners.
- Designed schema evolution and data partitioning in Redshift/Snowflake to optimize regulatory queries.
- Applied field-level encryption and PII/PCI masking for data privacy.
- Built Power BI dashboards and automated reporting for compliance, operations, and finance teams.
- Automated deployment using Jenkins, Docker, and Git for all data workflows.
- Modernized the data warehouse with Spark and Redshift Spectrum to support real-time analytics.
- Enabled monitoring and anomaly detection with AWS CloudWatch, ensuring reliable fintech operations.
- Conducted root-cause analysis for high-severity production issues and implemented long-term fixes.
- Performed continuous optimization of SQL queries to support compliance audit cycles.
- Created documentation for data models, ETL workflows, and operational playbooks.
- Collaborated with InfoSec to conduct regular security assessments on cloud data assets.
- Integrated REST API data sources from fintech partners, handling schema drift and data normalization.
- Supported UAT and production validation by building automated data quality checks and alerting.

Technomic

Jul 2015 – Feb 2018

Senior Data Engineer

- Built ETL workflows using Informatica and Python for claims, billing, and provider network data.
- Migrated on-prem data warehouses to Azure Data Lake and Synapse for improved analytics and compliance.
- Enforced HIPAA, SOC2, and HITRUST standards with data masking, audit logging, and secure reporting.
- Developed streaming data pipelines using Spark and Kafka for real-time clinical and operational feeds.
- Managed metadata and data lineage documentation for compliance and governance.
- Automated BI dashboard delivery using Power BI and Tableau for business stakeholders.
- Conducted anomaly detection and cleansing for high-velocity healthcare datasets.
- Integrated AI-powered enrichment for text and code normalization in medical records.
- Supported DevOps practices with Git, Jenkins, and Terraform for production-grade deployments.
- Built scheduling and workflow automation using Airflow and Azure Data Factory.
- Designed and executed data migration strategies with minimal disruption to business operations.
- Performed feature engineering and dimensional modeling for analytics and reporting.
- Coordinated with business analysts to translate requirements into technical solutions.
- Tuned Spark and SQL workloads for large-scale, distributed healthcare data processing.
- Authored operational runbooks and knowledge base articles for production data support.

EDUCATION

Bachelor of Science in Computer Science, University of Central Florida

Orlando, FL, May 2015